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SECTION 1

The AI Wild West: prototype vs. production vs. sprawl

- **The demo trap:** A two-week prototype that wows leadership rarely survives concurrent users, cost spikes, or compliance review without months of retrofitting.
- **Sam's wake-up:** When Sam's fintech chatbot went live, unpredictable latency, outage under a marketing blast, invisible spend, hallucinated policy advice, and stale FAQ content exposed gaps no single API call could close.
- **Prototype vs. production:** Table contrasts tolerate sloppy response times in demos with SLO-grade performance, budgeting, observability, grounded knowledge, and systematic experimentation in production.
- **Shared services counterfactual:** Observability, queuing, cost controls, guardrails, live data, and evaluation—if they already existed—would have let Sam ship features instead of plumbing.
- **AI sprawl:** When every team builds its own session store, key handling, and logging, security audits fail, lessons do not travel, and new AI work slows even though local engineering choices were sound.

SECTION 2

The model is a small slice of the system

- **2015 ML iceberg:** The classic NeurIPS debt paper showed model code as a tiny fraction of traditional ML systems surrounded by data, serving, and monitoring.
- **2026 GenAI shift:** A three-line chat completion might be ~2% of effort while knowledge pipelines, safety layers, sessions, tools, observability, routing, and orchestration consume the rest.
- **Same use case, new rings:** Customer support moved from a loaded classifier tensor to retrieval, multi-turn memory, tool calls, and governance—complexity exploded while surface code stayed short.
- **Why Sam's rewrite hurt:** The chatbot was never "just OpenAI"; it was a distributed application whose missing rings only appear when users and lawyers show up.

SECTION 3

First principles: what AI applications need

- **Frameworks vs. patterns:** LangChain-style tools help, but understanding core requirements improves debugging, reduces lock-in, and highlights where custom pieces belong.
- **Context-aware intelligence:** Applications must fuse live organizational knowledge with conversational state, not one-off prompts frozen at deploy time.
- **Multi-step orchestration:** Real workflows chain OCR, policy checks, approvals, and integrations—any step can fail and needs coordinated recovery.
- **Dynamic tools and safety:** Assistants must call external systems with quotas and auth while enforcing compliance on every input, output, and tool argument—not only via prompt hope.
- **Observability, experimentation, abstraction, scale:** Production demands traces and A/B infrastructure, provider-neutral model access, and infrastructure that grows without rewriting apps when load jumps from fifty to thousands of users.

SECTION 4

The platform blueprint

- **One concern per service:** Data plus Session deliver context; Workflow orchestrates; Tools integrate; Guardrails enforce policy; Observability and Evaluation instrument; Model Service hides vendors; Infrastructure and API Gateway carry cross-cutting ops.
- **Requirement mapping:** Each first-principle need maps to a dedicated service so teams implement business logic while inheriting shared behavior.
- **Independent scaling:** Clear boundaries let high-traffic features scale without dragging unrelated apps through coordinated redeploys.
- **Teaching implementation:** The book's platform is both reference architecture and runnable code teams can adopt whole or in part.

SECTION 5

Platform in action: one conversational request

- **Maria's question:** A retail shopper asking about electronics returns triggers a path that is simple in the UI but multi-service behind the scenes.
- **Workflow orchestration:** The Workflow Service loads session history (laptop interest), validates input and output via Guardrails, retrieves policy chunks from Data, and invokes Model generation with grounded context.
- **Session persistence:** The exchange is stored so a follow-up about opened packaging does not require repeating prior context.
- **Observability by default:** Token counts, timings, and per-service touches are recorded without bespoke instrumentation in application code.
- **Developer focus:** Application authors shape customer experience while the platform hides distributed coordination.

SECTION 6

Why platform thinking changes the organization

- **Consolidate repeated work:** Session handling, cost tracking, and auth become platform capabilities instead of parallel one-off implementations.
- **Consistent security:** Guardrails and infrastructure defaults make secure patterns the easy path rather than hoping each team nails key storage and sanitization.
- **Shared learning:** Prompt wins and safety configs can flow into platform services and evaluation tooling so optimization is organizational, not tribal.
- **Vendor flexibility:** Model abstraction lets teams adopt new providers or models without rewriting every workflow when pricing or capability shifts.
- **Book roadmap:** Subsequent chapters each deepen one service so readers can build production-grade AI estates that avoid Sam's cautionary arc.